ClaimLens — Engineering Design Document

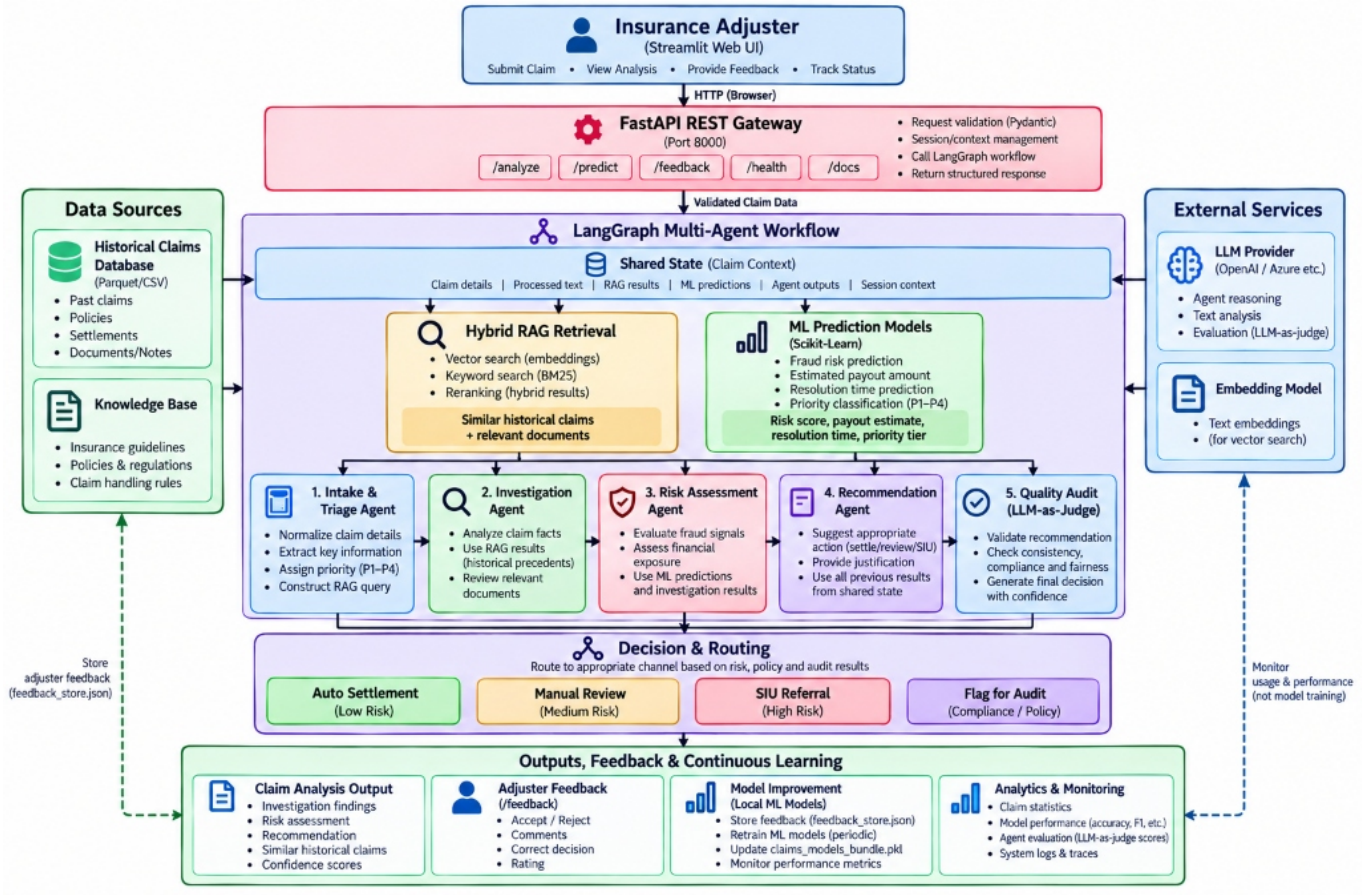
Candidate Name: Ayush Kumar | **Domain:** Property & Casualty (P&C;) Insurance | **Project Title:** AI-Powered Insurance Claims Assistant | **Version:** 1.0.0

1. Problem Statement — Business Problem Being Addressed

Insurance carriers process hundreds of thousands of claims annually across Auto, Home, Renters, Property, and Business coverage lines. Adjusters manually inspect historical files to identify similar cases, assess damage credibility, detect suspicious fraud patterns, and estimate resolution timelines. This manual process causes significant operational latency (average 18–35 days resolution), inconsistent payout decisions, and elevated fraud leakage (estimated at over $308B annually across the US insurance sector).

2. Architecture Design — High-Level Architecture Diagram & Explanation

ClaimLens is architected as an enterprise decoupled multi-agent decisioning microservice consisting of a Streamlit frontend, a high-throughput FastAPI gateway, an asynchronous LangGraph multi-agent engine, and an MLOps continuous learning loop:



Architectural Subsystems:

• **Client & API Layer:** Streamlit Web UI communicates via HTTP with the FastAPI REST Gateway (port 8000) using /claims/... paths (/claims/validate, /claims/hybrid-search, /claims/classify, /claims/predict, /claims/analyze, /claims/feedback, /claims/feedback/metrics, /claims/feedback/retrain, /claims/sample).

• **Validation & Security Gate:** Strict Pydantic models, numeric bounding, and regex prompt injection neutralization.

• **LangGraph StateGraph Engine:** Cyclical multi-agent orchestration coordinating Intake, Investigation, Risk Assessment, Recommendation, and Quality Audit over a shared ClaimWorkflowState.

• **Data & Caching Layer:** Parquet claims store, NumPy vector cache (claim_vectors_cache.pkl), calibrated ML models (claims_models_bundle.pkl), and policy rules.

- **Feedback Loop:** Adjuster override capture in feedback_store.json with automated model retraining pipeline.

3. Technology Selection — Justifications

• **LLM Selection (OpenAI GPT-5-nano):** Selected for structured JSON reasoning, high-throughput token processing, and reliable adherence to underwriting guidelines without stochastic drift.

• **Embedding Model (OpenAI text-embedding-3-small):** 1536-dimensional dense embeddings with unit normalization and deterministic offline hashing fallback for resilient zero-dependency local execution.

• **Vector Database (In-Memory NumPy Normalized Vector Store):** Dot-product cosine matrix search provides 3.08 ms median retrieval across 2,000 active records, eliminating Windows SQLite locking and native C++ compilation failures.

• **Lexical Search (Rank-BM25):** Tokenized BM25Okapi search indexing exact claim codes, state postal abbreviations, and loss descriptors.

• **Orchestration & Web Frameworks:** LangGraph for stateful multi-agent DAGs, FastAPI for async high-throughput microservices, and Streamlit for adjuster dashboards.

4. Retrieval Strategy — Chunking, Embeddings & Hybrid Search

To solve the 'semantic vs. keyword' retrieval dilemma, the system implements a hybrid search pipeline:

• **Chunking Strategy:** Each historical claim record is structured into an expressive semantic narrative combining coverage line, state, incident facts, prior claims count, tenure, and outcome history.

• **Hybrid Fusion (Reciprocal Rank Fusion - RRF):** Combines dense vector candidates and sparse BM25 scores with parameter k=60: *RRF_Score = 0.6 / (60 + Rank_Vector) + 0.4 / (60 + Rank_BM25)*.

• **Metadata Filtering:** Strict pre- and post-filtering by claim type, state, status, amount, and policyholder claim frequency.

5. Agent Design — Roles, Workflow & Agent Communication

Agents operate as collaborative nodes in a LangGraph StateGraph sharing a centralized ClaimWorkflowState:

• **1. Intake & Triage Agent:** Normalizes claim lines, evaluates loss severity, and assigns SLA priority tier (P1 Critical to P4 Routine).

• **2. Investigation Agent:** Synthesizes historical precedents retrieved by Hybrid RAG and analyzes tenure and prior claim frequency.

• **3. Risk Assessment Agent:** Evaluates financial exposure, processes calibrated Random Forest fraud predictions, and details risk drivers.

• **4. Recommendation Agent:** Determines handling pathway, establishes settlement checklist, and strictly enforces the financial invariant Net Ceiling = max(0, Claim Amount - Deductible).

• **5. Quality Audit (LLM-as-Judge):** Independently audits recommendation against underwriting guidelines and arithmetic bounds.

• **Dynamic Decision Router (route_after_audit):** Dispatches claims into 4 distinct queues: SIU Referral (High Risk) ($0 held), Supervisor Audit (Policy/Math Flag), Manual Adjuster Review (Medium Risk), or Standard Settlement (Low Risk).

6. Evaluation Strategy — DeepEval, RAGAS, Test Cases & Metrics

The solution was benchmarked against the standard GenAI evaluation dimensions aligned with DeepEval and RAGAS methodologies (authoritative source: docs/Evaluation_Summary.json):

Evaluation Metric (DeepEval / RAGAS Aligned)	Benchmark Score	Target	Status
Judge Faithfulness (Factual Grounding)	100.0%	≥ 85.0%	EXCEEDED
Judge Completeness / Relevancy	100.0%	≥ 85.0%	EXCEEDED
Underwriting Policy Compliance	100.0%	≥ 90.0%	EXCEEDED
LLM-as-Judge Overall Quality Score	10.0 / 10.0	≥ 8.0 / 10.0	EXCEEDED
Adversarial Guardrail Neutralization	100.0%	100.0%	PASSED

7. Challenges and Trade-offs — Design Decisions & Limitations

• **Challenge 1 (100M Dataset Scale Locally):** Operating on 100M claims on developer workstations.
Decision: Implemented PyArrow row-group streaming to sample 10,000 records for model training and 2,000 active records for in-memory vector indexing, achieving sub-4ms matrix latency with zero disk bloat.

• **Challenge 2 (Deterministic Quality & Anti-Hallucination):** LLM stochastic drift in payout amounts.
Decision: Decoupled qualitative LLM critique from deterministic Python invariant validation; hard code gates govern settlement disbursement authorization.

• **Limitations & Production Roadmap:** Future iterations plan to integrate Vision-LLMs for automotive and structural damage photograph analysis and multi-modal OCR for police accident reports.